Notes

Sample Mean, Sample Variance, and Unbiased Estimation

Based on Sheldon M. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 5th ed., Section 6.4

1. Population and sample: notation and formulas

A **population** is the complete collection of interest. A **sample** is a collection of observations drawn from that population. Population parameters are fixed quantities; sample statistics depend on the observations selected.

Quantity	Population	Sample
Number of observations	N	n
Mean	$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Variance	$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$	$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$
Standard deviation	σ	s

The sample formula s^2 is designed to estimate the population variance σ^2 when the population mean μ is unknown. The precise unbiasedness argument below assumes independent, identically distributed observations with finite variance.

2. Why is there one fewer degree of freedom?

The deviations from the sample mean always sum to zero:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - n\bar{x} = 0.$$

For example, with $n = 3$ and $\bar{x} = 6$, we have $x_1 + x_2 + x_3 = 18$. If $x_1 = 4$ and $x_2 = 5$, then $x_3 = 9$ is determined. Similarly, once $n - 1$ deviations are fixed, the final deviation is fixed by their sum being zero. Thus the deviations have $n - 1$ **degrees of freedom**.

However, degrees of freedom alone is an intuition. The next calculation demonstrates exactly why division by $n - 1$ gives an unbiased estimator.

3. Derivation of the $n - 1$ denominator

Let $X_1, \ldots, X_n$ be independent and identically distributed observations satisfying

$$E[X_i] = \mu, \qquad \mathrm{Var}(X_i) = \sigma^2.$$

Define the sample mean and sample variance as

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \quad n \geq 2.$$

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First expand the squared deviations:

$$\begin{aligned}\sum_{i=1}^{n}(X_i-\bar{X})^2 &= \sum_{i=1}^{n}(X_i^2-2X_i\bar{X}+\bar{X}^2)\\ &= \sum_{i=1}^{n}X_i^2-2\bar{X}\sum_{i=1}^{n}X_i+n\bar{X}^2\\ &= \sum_{i=1}^{n}X_i^2-2n\bar{X}^2+n\bar{X}^2\\ &= \sum_{i=1}^{n}X_i^2-n\bar{X}^2.\end{aligned}$$

We also know, from the definition of variance, that

$$E[X_i^2]=\sigma^2+\mu^2.$$

Since the observations are independent,

$$E[\bar{X}]=\mu, \qquad \mathrm{Var}(\bar{X})=\frac{\sigma^2}{n}, \qquad E[\bar{X}^2]=\frac{\sigma^2}{n}+\mu^2.$$

Taking expectations of the expanded squared-deviation identity gives

$$\begin{aligned}E\left[\sum_{i=1}^{n}(X_i-\bar{X})^2\right] &= \sum_{i=1}^{n}E[X_i^2]-nE[\bar{X}^2]\\ &= n(\sigma^2+\mu^2)-n\left(\frac{\sigma^2}{n}+\mu^2\right)\\ &= n\sigma^2+n\mu^2-\sigma^2-n\mu^2\\ &= (n-1)\sigma^2.\end{aligned}$$

Therefore,

$$E[S^2]=\frac{1}{n-1}E\left[\sum_{i=1}^{n}(X_i-\bar{X})^2\right]=\frac{(n-1)\sigma^2}{n-1}=\boxed{\sigma^2}.$$

Conclusion: S^2 is an *unbiased estimator* of population variance because $E[S^2]=\sigma^2$. The denominator $n-1$ corrects the average downward bias introduced by measuring deviations from the estimated sample mean $\bar{X}$.

4. What happens if we divide by n?

Define $V_n=\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X})^2$. Then

$$E[V_n]=\frac{n-1}{n}\sigma^2\neq\sigma^2.$$

For example, when $n=5$, $E[V_5]=\frac{4}{5}\sigma^2$: on average this estimator is 20% below the true population variance. This does *not* imply that the $n-1$ result is closer to the true value in every individual sample.

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5. Known mean versus unknown mean

Situation	Variance formula	Divisor
Entire finite population	$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$	N
Random sample, true mean μ known	$\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$	n
Random sample, mean estimated by $\bar{X}$	$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$	$n-1$

For the second row, the estimator is unbiased because $E[(X_i-\mu)^2] = \sigma^2$ when μ is the true population mean. For the third row, the estimator is unbiased under the i.i.d. sampling assumption used above.

One-sentence explanation: Estimating μ using $\bar{X}$ makes the deviations around the fitted centre systematically too small in squared sum; dividing by $n-1$ rather than n corrects this in expectation.

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6. Standardization of an observation and of a sample mean

A Z-score measures how far a value lies from its mean in units of its standard deviation. The same idea is used for an individual observation and for a sample average. The denominators differ because those two random quantities have different variances.

6.1 One observation

Let X have population mean μ and population variance σ^2. Its standard deviation is $\sigma = \sqrt{\sigma^2}$. Subtracting the mean centres X at zero; dividing by its standard deviation makes the variance equal to one:

$$\begin{aligned} Z &= \frac{X-\mu}{\sigma}, \\ E[Z] &= \frac{E[X]-\mu}{\sigma} = 0, \\ \mathrm{Var}(Z) &= \frac{\mathrm{Var}(X)}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1. \end{aligned}$$

If $X \sim N(\mu, \sigma^2)$, then $Z \sim N(0,1)$ exactly. The standard normal table gives $\Phi(z) = P(Z \le z)$. This is the appropriate transformation when a probability concerns *one observation*.

6.2 The average of n observations

Let $X_1, \ldots, X_n$ be independent observations with the same mean μ and variance σ^2. Define

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

The mean and variance of the average follow directly from the rules for expectation and variance:

$$\begin{aligned} E[\bar{X}] &= \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{n\mu}{n} = \mu, \\ \mathrm{Var}(\bar{X}) &= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\ &= \frac{1}{n^2}\,\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) \qquad \text{(independence)} \\ &= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. \end{aligned}$$

Variance has squared units, whereas standardization requires a *standard deviation*. Taking the square root gives

$$\mathrm{SD}(\bar{X}) = \sqrt{\mathrm{Var}(\bar{X})} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}.$$

This quantity is called the *standard error of the sample mean* when σ is known. Therefore its standardized form is

$$\boxed{Z_{\bar{X}} = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}.$$

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The denominator is $\sigma/\sqrt{n}$, **not** σ/n. The latter would not give unit variance:

$$\mathrm{Var}\left(\frac{\bar{X}-\mu}{\sigma/n}\right) = \frac{\sigma^2/n}{\sigma^2/n^2} = n.$$

If the X_i are normally distributed, this sample-mean Z-score is exactly standard normal for any n. If the parent distribution is not normal, the central limit theorem often gives an approximately standard normal Z-score for sufficiently large n; the quality of the approximation depends on the parent distribution and sampling conditions.

6.3 Which formula should be used?

	Single observation	**Sample average**
Random quantity	X	$\bar{X} = \sum X_i/n$
Expected value	μ	μ
Variance	σ^2	σ^2/n
Standard deviation	σ	$\sigma/\sqrt{n}$
Standardization	$\dfrac{X-\mu}{\sigma}$	$\dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}}$
Typical question	One randomly selected value	Average of n independently selected values

In either case, standardization means subtracting the appropriate mean and dividing by the appropriate standard deviation. The numerator has the same structure; it is the standard deviation in the denominator that changes.

6.4 Numerical example: an individual mark and an average mark

Assume individual examination marks are normally distributed with mean 60 and variance 100. Thus $\sigma = 10$.

One randomly selected student. For $X \sim N(60, 100)$,

$$\begin{aligned} P(X<70) &= P\left(\frac{X-60}{10} < \frac{70-60}{10}\right) \\ &= P(Z<1) = \Phi(1) \approx \boxed{0.8413}. \end{aligned}$$

Average of 25 independently selected students. Here $n = 25$, so

$$E[\bar{X}] = 60, \qquad \mathrm{Var}(\bar{X}) = \frac{100}{25} = 4, \qquad \mathrm{SD}(\bar{X}) = \sqrt{4} = 2.$$

Consequently,

$$\begin{aligned} P(\bar{X}<70) &= P\left(\frac{\bar{X}-60}{2} < \frac{70-60}{2}\right) \\ &= P(Z<5) = \Phi(5) \approx \boxed{0.9999997}. \end{aligned}$$

The two probabilities differ because an average of 25 independent marks fluctuates much less than a single mark, even though both have expected value 60.

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6.5 Connection with sample variance and the $n-1$ divisor

The notation $\sigma/\sqrt{n}$ comes from the *variance of a sample mean*, $\mathrm{Var}(\bar{X}) = \sigma^2/n$. It is a separate result from the $n-1$ correction used to estimate an *unknown population variance*:

$$\underbrace{\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}}_{\text{variability of the sample mean}} \quad , \quad \underbrace{S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}_{\text{estimate of population variance}}.$$

The first describes how sample averages vary across repeated samples. The second describes how individual observations vary within a sample and corrects for estimating μ using $\bar{X}$. If σ is unknown, the estimated standard error is $S/\sqrt{n}$. For a normal population, substituting S for σ changes the reference distribution from standard normal to Student's t distribution with $n-1$ degrees of freedom; this is not an exact Z statistic.

Practical rule: Standardize the quantity actually asked about. Use σ for one observation X; use $\sigma/\sqrt{n}$ for the average $\bar{X}$ of n independent observations. Neither standardization formula should be confused with the $n-1$ divisor used in the unbiased sample variance.

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